

UML Design Diagrams

Campus Grade

AI-Powered Assignment Creation, Submission and Intelligent Evaluation Platform

Class Diagram • Component Diagram • Deployment Diagram • Use Case Diagram

Prepared by: Yashika

Roll Number: 2501560028

Department of Computer Applications (MCA)

K.R. Mangalam University, Sohna, Gurugram, Haryana

1. Class Diagram

The class diagram models the core entities of the Campus Grade platform: User (with Student, Faculty, and Admin specializations), Assignment and Submission, the AI Evaluation Engine and Plagiarism Checker, and the resulting EvaluationResult / GradeRecord, alongside the persistence and notification layers. It captures key attributes, operations, and associations — including the generalization of Student/Faculty/Admin from User and the composition between an EvaluationResult and its GradeRecord.

Class Diagram — Campus Grade: AI-Powered Assignment Creation, Submission & Intelligent Evaluation Platform

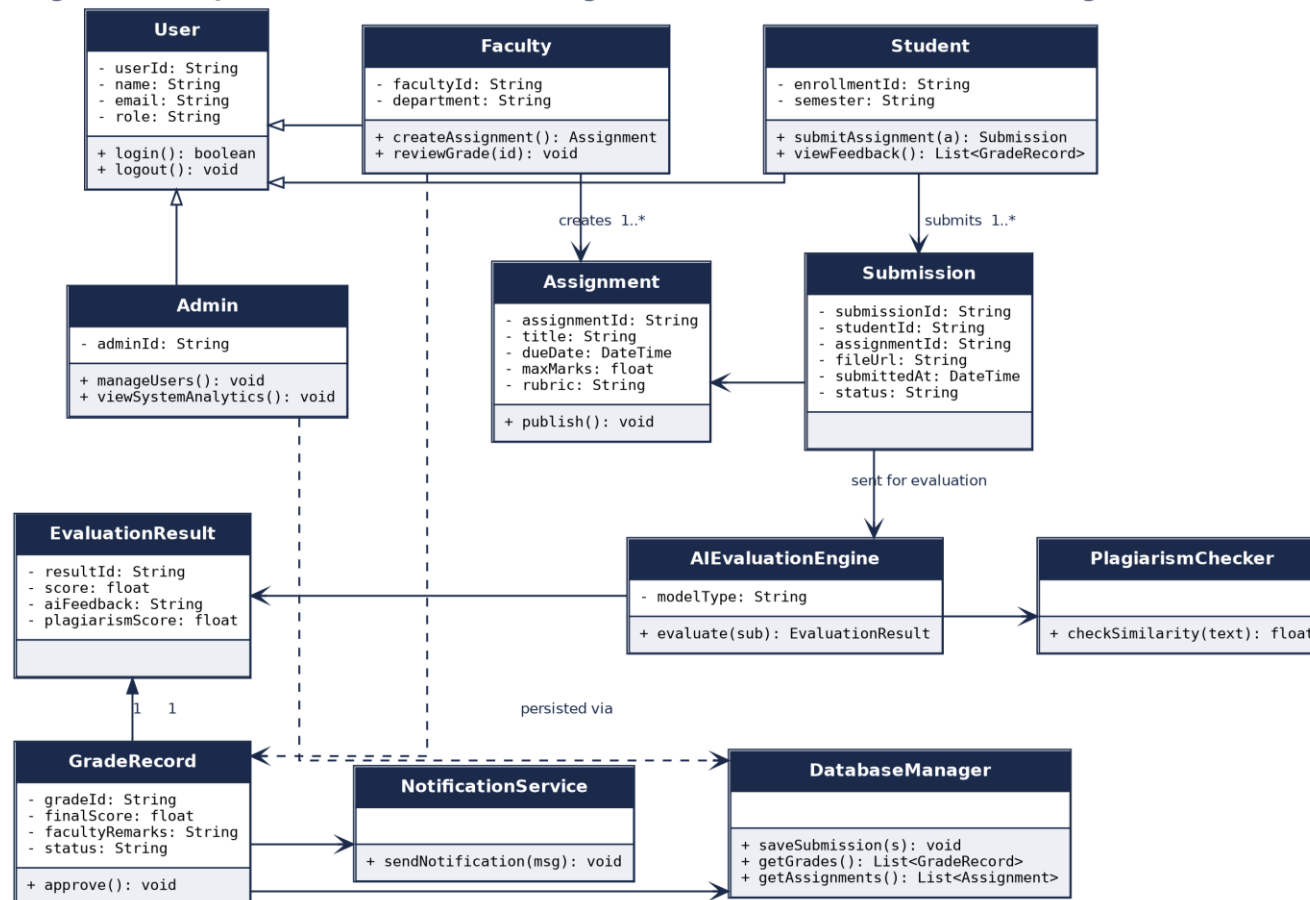


Figure 1: Class Diagram

2. Component Diagram

The component diagram decomposes the platform into deployable, interacting units: the Student and Faculty Portals (Presentation Layer); the Backend API, which stores submitted files in Cloud File Storage and triggers evaluation; the AI Evaluation Pipeline (AI Evaluation Engine, Plagiarism Detection, and Grading & Feedback); and the Database, which persists grades before the Notification component alerts the student.

Component Diagram — Campus Grade: AI-Powered Assignment Creation, Submission & Intelligent Evaluation Platform

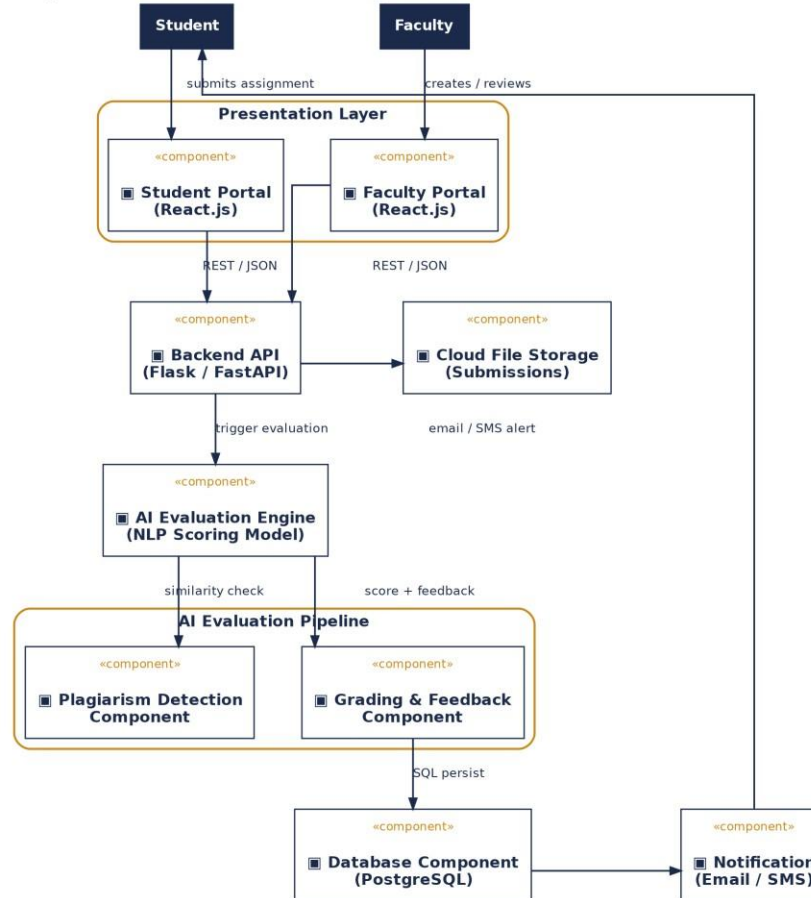


Figure 2: Component Diagram

3. Deployment Diagram

The deployment diagram maps the platform's software artifacts onto physical/virtual nodes: the Client Workstation/Browser (React.js Web Portal used by students and faculty) communicates over HTTPS/REST with the Application Server (Flask/FastAPI). The Application Server delegates evaluation requests to the GPU-enabled AI/ML Processing Server, persists data to the Database Server (PostgreSQL) over TCP, and stores assignment/submission files on the Cloud Storage Server over HTTPS.

Deployment Diagram — Campus Grade: AI-Powered Assignment Creation, Submission & Intelligent Evaluation Platform

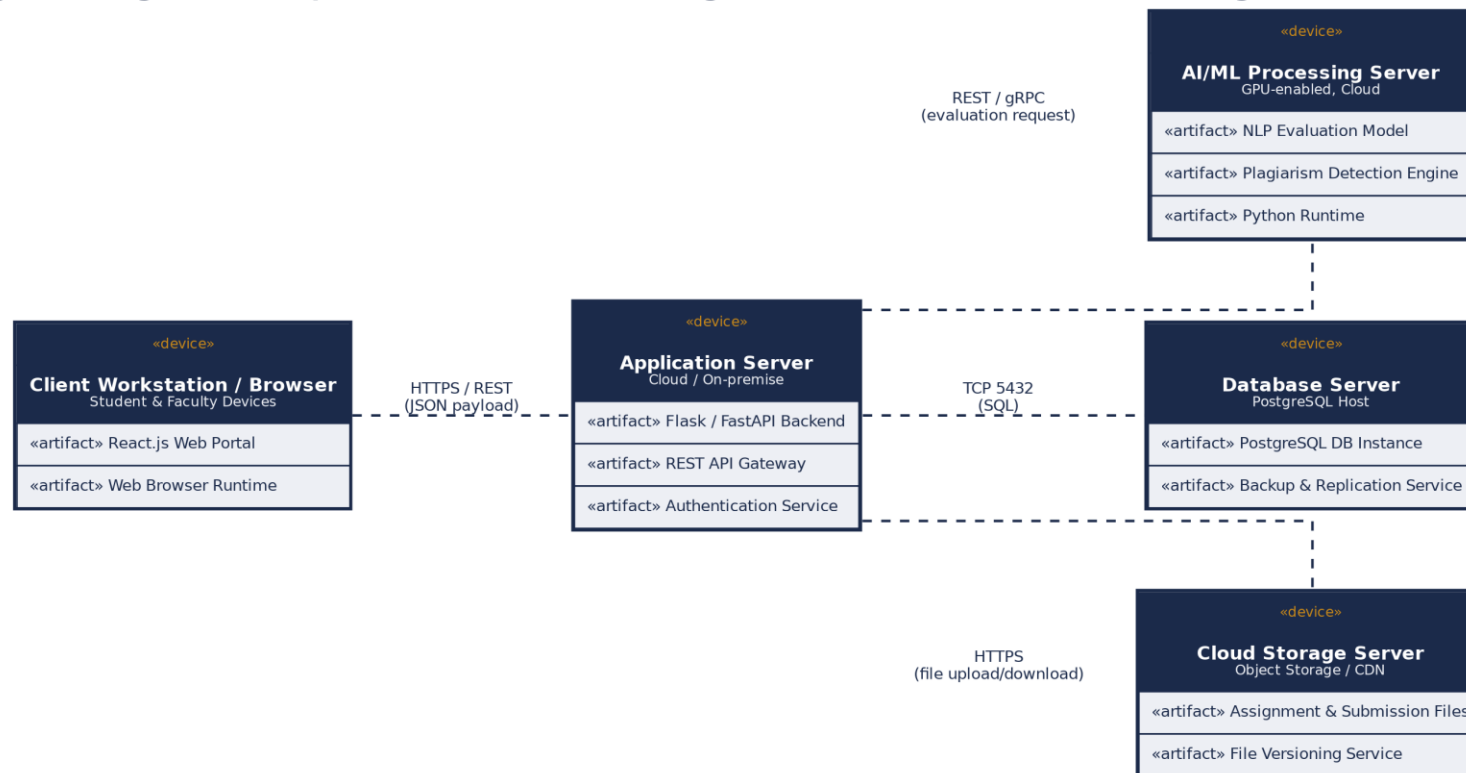


Figure 3: Deployment Diagram

4. Use Case Diagram

The use case diagram identifies three actors — Student, Faculty, and Admin — and the primary use cases they interact with. The submission-to-grading pipeline (Submit Assignment, Upload Submission File, Auto-Evaluate Submission, Check Plagiarism, Generate Score & Feedback, Review & Approve Grade) is chained through «include» relationships representing the automated AI evaluation flow, while Faculty create/publish assignments and review grades, and Admin manages users, reports, and platform login.

Use Case Diagram — Campus Grade: AI-Powered Assignment Creation, Submission & Intelligent Evaluation Platform

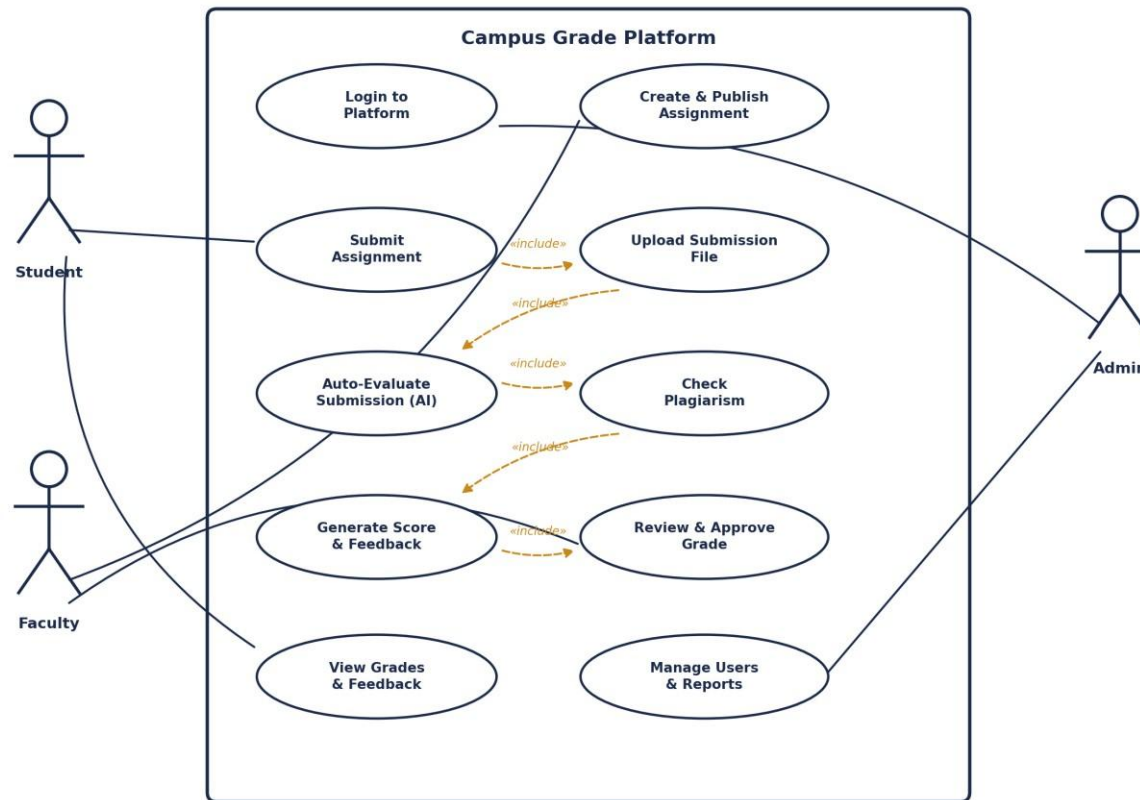


Figure 4: Use Case Diagram